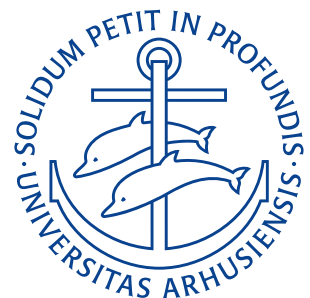


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Exercises

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Date: April 20, 2026



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Lecture 1: Introduction

17. Marts 2026

Problem 1.1

A laser printer uses a laser beam to print copy rapidly for a computer. The laser is positioned by a control input $r(t)$, so that we have

$$Y(s) = \frac{4(s+50)}{s^2+30s+200} R(s).$$

The input $r(t)$ represents the desired position of the laser beam.

- If $r(t)$ is a unit step input, find the output $y(t)$.
- What is the final value of $y(t)$?

Lecture 2: Mathematical Models of Systems

20. Marts 2026

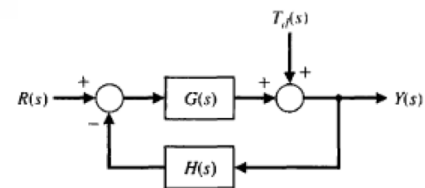
Lecture 3: Feedback Control System Characteristics

24. March 2026t

Problem E4.11

Consider the closed-loop system in the figure, where

$$G(s) = \frac{K}{s+10}, \quad H(s) = \frac{14}{s^2+5s+6}.$$



- Compute the transfer function $T(s) = Y(s)/R(s)$.
- Define the tracking error to be $E(s) = R(s) - Y(s)$. Compute $E(s)$ and determine the steady-state tracking error due to a unit step input, that is, let $R(s) = 1/s$.
- Compute the transfer function $Y(s)/T_d(s)$ and determine the steady-state error of the output due to a unit step disturbance input, that is, let $T_d(s) = 1/s$.
- Compute the sensitivity S_k^T

Lecture 4: Performance of Feedback Control Systems

27. March 2026

Lecture 5: Stability of Linear Feedback Systems

7. April 2026

Lecture 5: Stability of Linear Feedback Systems

7. April 2026

Lecture 6: Root Locus Method, I

9. April 2026

Lecture 7: Root Locus Method, II

14. April 2026